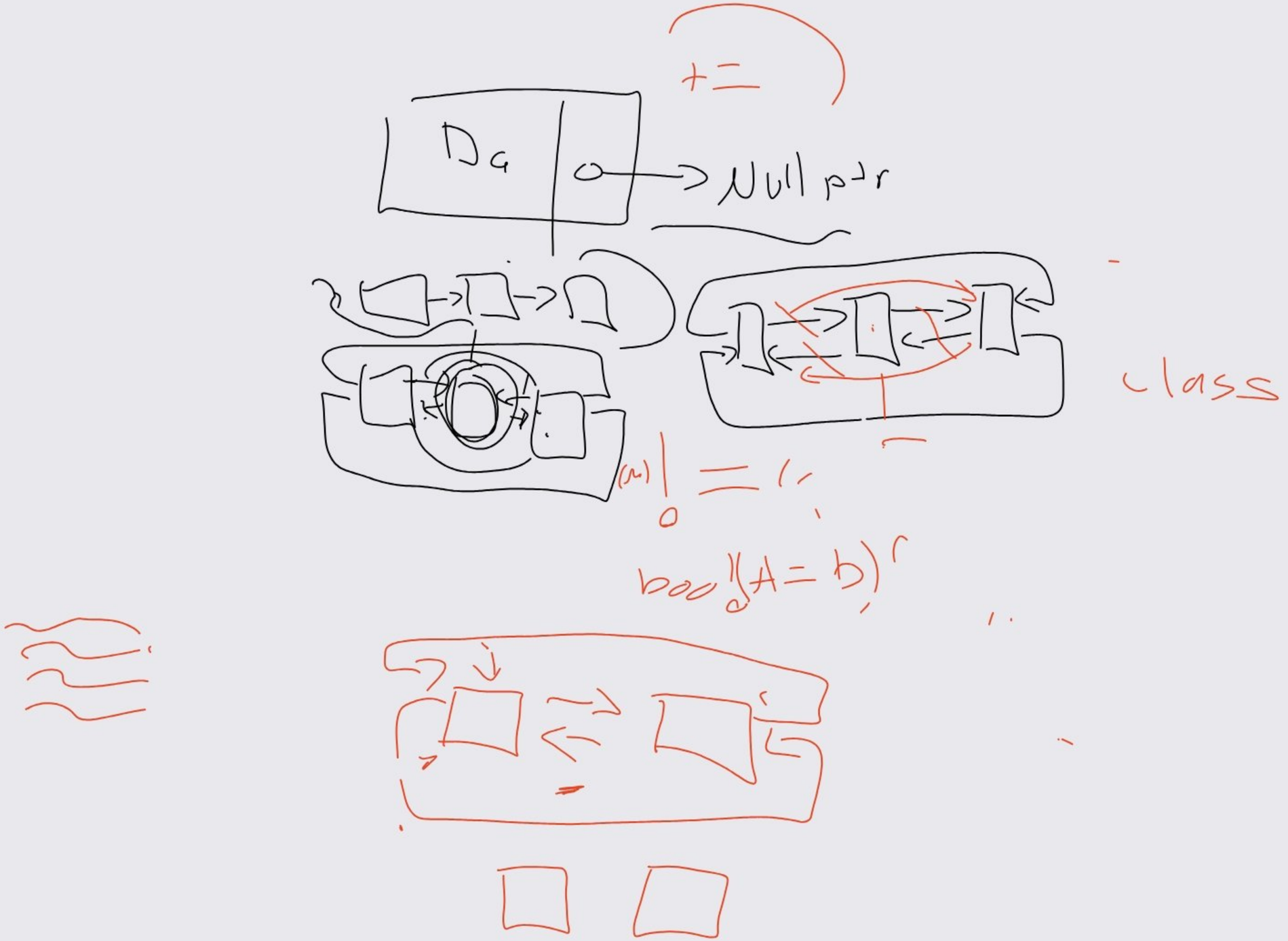
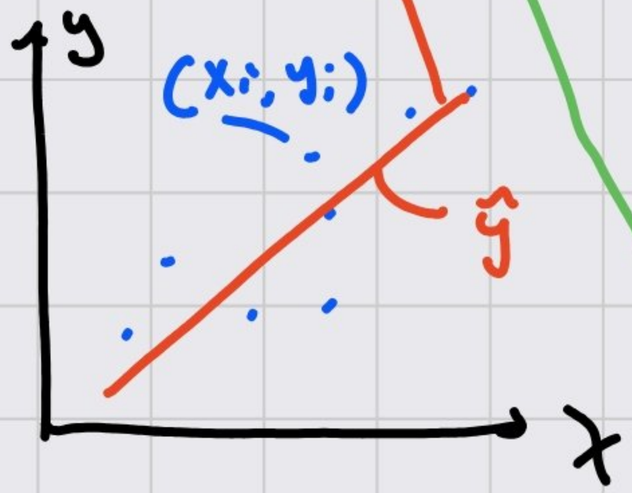

+=
Dq
Null ptr
class

- Regresión Lineal Simple

Modelo

$$\hat{y} = \beta_0 + \beta_1 X$$



Función de Pérdida / Error

$$\mathcal{L}(\vec{\beta}) = \frac{1}{n}\sum(\hat{y}_i - y_i)^2$$

residuos → $\varepsilon_i = \hat{y}_i - y_i$

$\min(\mathcal{L})$ $\quad \nabla\mathcal{L} = 0$

Parámetros óptimos

$\beta_1 = \frac{\sigma_{xy}}{\sigma_x^2}$ Analítica

$\beta_0 = \bar{y} - \beta_1 \bar{x}$

Evaluación

Métricas → R^2 $[0,1]$, MSE, RMSE, MAE, MAPE

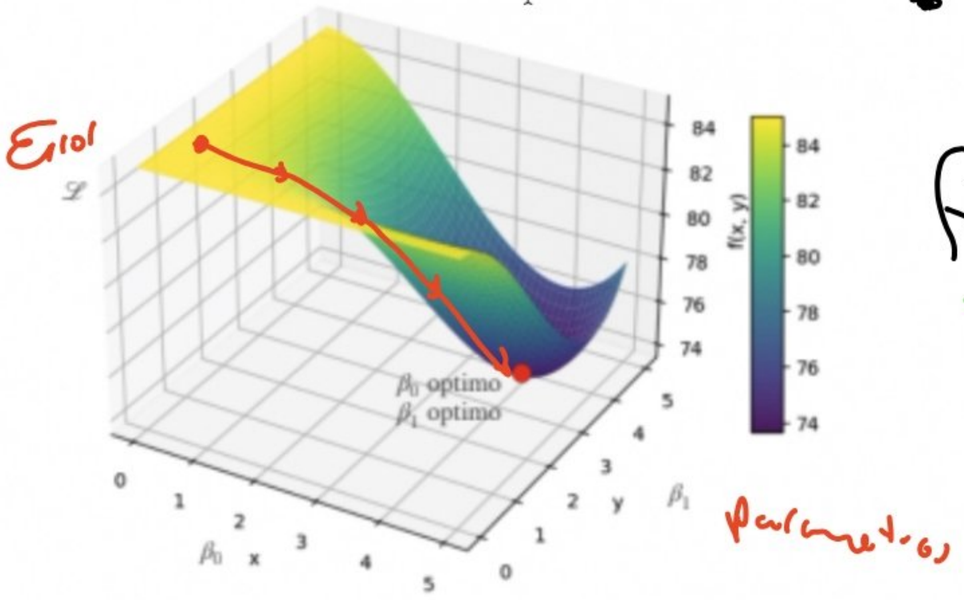
> Residuos $\varepsilon_i \sim N(0, \sigma_\varepsilon^2)$

Supernova remnant G46.8- | Problem loading page | Sesión 1 - Repaso de M | t.dvi - minka-matrix.pdf | Reg_Mul_G1.ipynb - Co | raw.githubusercontent.com | numpy add column of c | hadamard product - Bu | Home | Prop_algebra
classroom.google.com/u/1/c/ODc0OTY0Njg0NjA4/m/ODc2MTI0MjA5ODg1/details
VPN

$$\mathcal{L} = \frac{1}{n}\sum_i (\hat{y}_i - y_i)^2 = \frac{1}{n}\sum ((\beta_0 + \beta_1 x_1 + \ldots + \beta_n x_n) - y_i)^2$$

$$\mathcal{L}(\vec{\beta})$$

$$\mathbb{R}^n \rightarrow \mathbb{R}$$



Parametros

- Analítica

$$\beta_1 = \frac{\sigma_{xy}}{\sigma_x^2} \qquad \beta_0 = \bar{y} - \beta_1 \bar{x}$$

- Aproximada (Ej., descenso del gradiente)

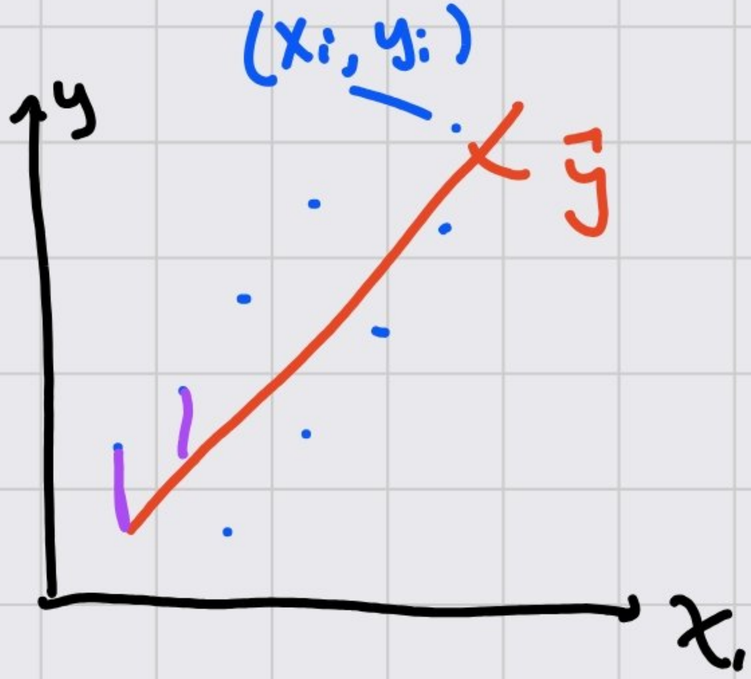
$$\beta_{n+1} = \beta_n - \alpha \nabla \mathcal{L}$$

- Regresión Lineal Múltiple

Modelo

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \ldots + \beta_m x_m$$

(x_1, x_2: characteristics, age, height, weight, ...; $m \rightarrow$ # de vars.)



Función de Pérdida

$$\mathcal{L}(\bar{\beta}) = \frac{1}{n}\sum(\hat{y}_i - y_i)^2$$

$$y_1 = \beta_0 + \beta_1 x_{11} + \beta_2 x_{12} + \ldots + \beta_m x_{1m} + \varepsilon_1$$

$$y_2 = \beta_0 + \beta_1 x_{21} + \beta_2 x_{22} + \ldots + \beta_m x_{2m} + \varepsilon_2$$

$$\vdots$$

$$y_n = \beta_0 + \beta_1 x_{n1} + \beta_2 x_{n2} + \ldots + \beta_m x_{nm} + \varepsilon_n$$

$$y_1 = \beta_0 + \beta_1 x_{11} + \beta_2 x_{12} + \dots + \beta_m x_{1m} + \varepsilon_1$$
$$y_2 = \beta_0 + \beta_1 x_{21} + \beta_2 x_{22} + \dots + \beta_m x_{2m} + \varepsilon_2$$
$$\vdots$$
$$y_n = \beta_0 + \beta_1 x_{n1} + \beta_2 x_{n2} + \dots + \beta_m x_{nm} + \varepsilon_n$$

$$\mathcal{L} = \frac{1}{n}\sum (\hat{y}_i - y_i)^2$$
$$\mathcal{L} = \frac{1}{n}\sum \varepsilon_i^2$$

Forma Matricial

$$\underset{n\times 1}{\begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}} = \underbrace{\begin{pmatrix} 1 & x_{11} & x_{12} \dots x_{1m} \\ 1 & x_{21} & x_{22} \dots x_{2m} \\ 1 & \vdots & \\ \vdots & & \\ 1 & x_{n1} & x_{n2} \dots x_{nm} \end{pmatrix}}_{n \times (m+1)} \underbrace{\begin{pmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_m \end{pmatrix}}_{(m+1)\times 1} + \underset{n\times 1}{\begin{pmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \varepsilon_n \end{pmatrix}}$$

intercepto

$$Y = X\beta + \varepsilon$$

TARGET — Matriz de características — params. — residuos

$$Y = XB + \varepsilon \Rightarrow \varepsilon = Y - XB$$

$$\varepsilon^T\varepsilon = (Y - XB)^T(Y - XB)$$

$$= (Y^T - X^TB^T)(Y - XB)$$

$$= Y^TY - Y^TXB - X^TB^TY + X^TB^TXB$$

$$\mathcal{L} = (\varepsilon^T \varepsilon) = Y^T Y - Y^T X\beta - X^T \beta^T Y + X^T \beta^T X\beta$$

$$\frac{\partial(\varepsilon^T \varepsilon)}{\partial \beta} = 0 = \frac{\partial}{\partial \beta}(Y^T Y) - \frac{\partial}{\partial \beta}(Y^T X\beta) - \frac{\partial}{\partial \beta}(X^T \beta^T Y)$$

$\frac{\partial}{\partial \beta}(Y^T Y) \to 0$

$\frac{\partial}{\partial \beta}(Y^T X\beta) \to Y^T X$

$\frac{\partial}{\partial \beta}(X^T \beta^T Y) \to X^T Y$

$\beta \to \beta$

$Y^T X = X^T Y$

$$+ \frac{\partial}{\partial \beta}(X^T \beta^T X\beta) \to 2X^T X\beta$$

$$= -Y^T X - X^T Y + 2X^T X\beta = 0$$

$$= -2X^T Y + 2X^T X\beta = 0$$

3×200 $\quad 200 \times 3 \to 3 \times 3$

$$X^T X\beta = X^T Y \Rightarrow \beta = (X^T X)^{-1} X^T Y$$

Parametros optimos

$$\beta = (\beta_0, \beta_1, \beta_2, \ldots \beta_3)$$

OLS Equation

Supernova remnant G46.8- | Problem loading page | Machine Learning e Inf | t.dvi - minka-matrix.pdf | raw.githubusercontent.com | numpy add column of | hadamard product - Bu | Home | Prop_algebra | Reg_Mult_G2.ipynb

colab.research.google.com/drive/1sH0aJZPyiKmhcemjfXjG7zuxZyOMM-25#scrollTo=dssO-TcQo2Hf

180% VPN

Next steps: Generate code with df | New interactive sheet

```
X = df[["TV","Radio","Newspaper"]].to_numpy()
X
```

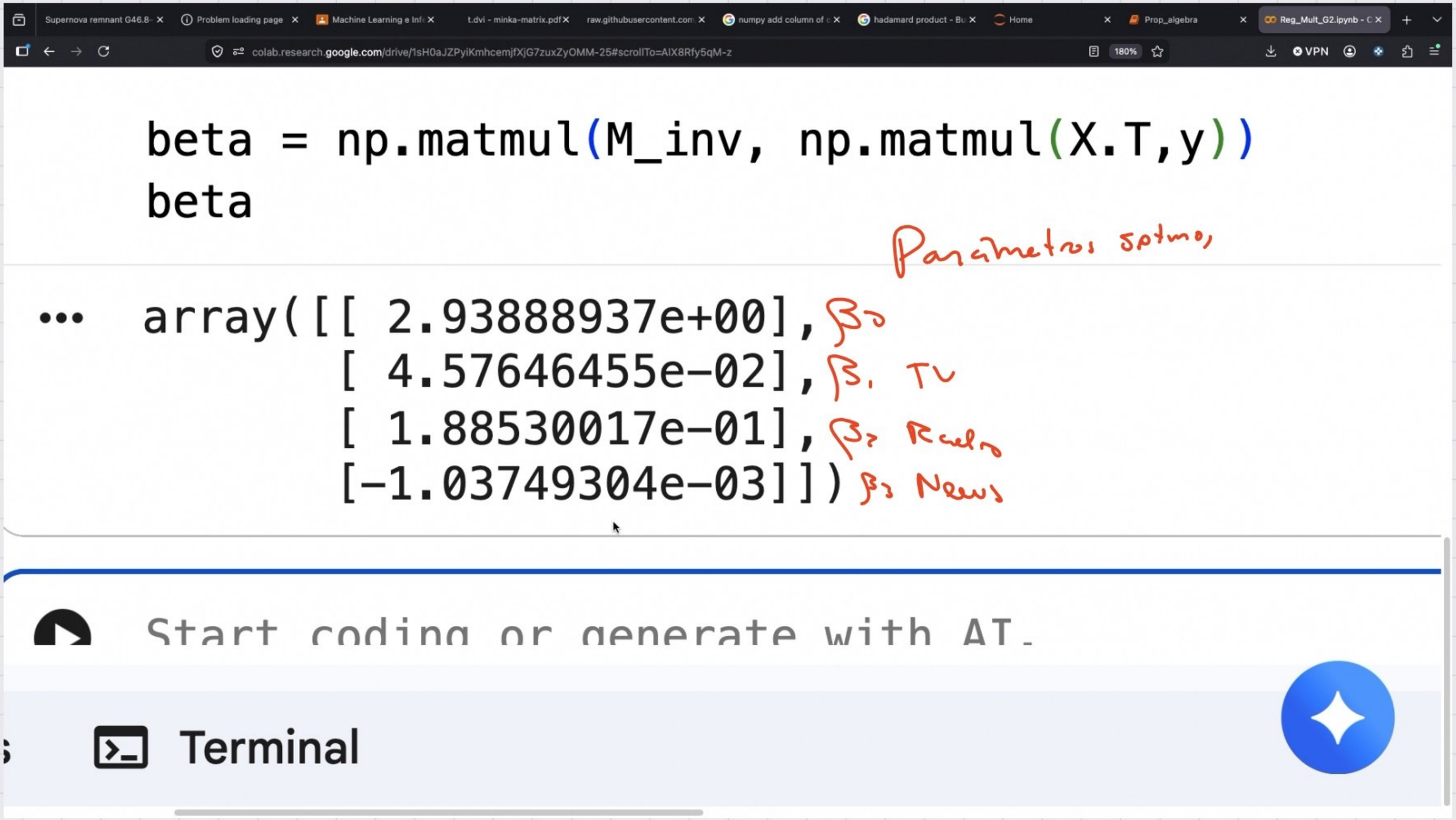
```
array([[230.1,  37.8,  69.2],
       [ 44.5,  39.3,  45.1],
       [ 17.2,  45.9,  69.3],
       [151.5,  41.3,  58.5],
       [180.8,  10.8,  58.4],
       [  8.7,  48.9,  75. ],
       [ 57.5,  32.8,  23.5],
       [120.2,  19.6,  11.6],
       [  8.6,   2.1,   1. ],
       [199.8,   2.6,  21.2],
       [ 66.1,   5.8,  24.2],
       [214.7,  24.       4. ]
```

Scalar Scalar = 8

Vector = np.array([8,5,3])

Matrix = ([[8,5,3],
[5,4,2],
[3,2,1]])

$$\begin{pmatrix} 8 & 5 & 3 \\ 5 & 4 & 2 \\ 3 & 2 & 1 \end{pmatrix}$$


Supernova remnant G46.8-
Problem loading page
Machine Learning e Inf
t.dvi - minka-matrix.pdf
raw.githubusercontent.com
numpy add column of
hadamard product - Bu
Home
Prop_algebra
Reg_Mult_G2.ipynb -
colab.research.google.com/drive/1sH0aJZPyiKmhcemjfXjG7zuxZyOMM-25#scrollTo=AIX8Rfy5qM-z
180%
VPN
beta = np.matmul(M_inv, np.matmul(X.T,y))
beta
Parameters optimal
array([[2.93888937e+00],
[4.57646455e-02],
[1.88530017e-01],
[-1.03749304e-03]])
β0
β1 TV
β2 Radio
β3 News
Start coding or generate with AI.
Terminal

Supernova remnant G46.8- | Problem loading page | Machine Learning e Inf | t.dvi - minka-matrix.pdf | raw.githubusercontent.com | numpy add column of | hadamard product - Bu | Home | Prop_algebra | Reg_Mult_G2.ipynb -
colab.research.google.com/drive/1sH0aJZPyiKmhcemjfXjG7zuxZyOMM-25#scrollTo=dssO-TcQo2Hf
180%
VPN

```
y = y.reshape(len(y),1)
```

```
# Parameters vector beta

M = np.matmul(X.T, X)

M_inv = np.linalg.inv(M)

beta = np.matmul(M_inv, np.matmul(X.T,y))
beta
```

$$\beta = (X^T X)^{-1} X^T Y$$

```
array([[ 2.93888937e+00],
       [ 4.57646455e-02],
       [ 1.88530017e-01],
       [-1.03749304e-03]])
```

Supernova remnant G46.8- | Problem loading page | Machine Learning e Inf | t.dvi - minka-matrix.pdf | raw.githubusercontent.com | numpy add column of | hadamard product - Bu | Home | Prop_algebra | Reg_Mult_G2.ipynb - C

colab.research.google.com/drive/1sH0aJZPyiKmhcemjfXjG7zuxZyOMM-25#scrollTo=-cdLuXrludqV

180% VPN

Reg_Mult_G2.ipynb

PRO

File Edit View Insert Runtime Tools Help

Share

Commands + Code + Text Run all

RAM Disk

$$\sum(\hat{y}_i - y_i)$$

$$R^2 = \frac{SSR}{SST} \quad [0,1]$$

$$MSE = \frac{1}{n}\sum(\hat{y}_i - y_i)^2$$

$$MAE = \frac{1}{n}\sum|\hat{y}_i - y_i|$$

$$RMSE = \sqrt{\frac{1}{n}\sum(\hat{y}_i - y_i)^2}$$

[36] ✓ 0s

```
# Paso 2 - X,y
X = df[["TV","Radio","Newspaper"]]
y = df["Sales"]

# Paso 3 - instanciar Modelo
from sklearn.linear_model import LinearRegression
model = LinearRegression()

# Paso 4 - Entrenar el modelo (con datos)
model.fit(X,y)

# Paso 5 - Predicciones (y_hat)
y_hat = model.predict(X)
```

[ ]

Start coding or generate with AI.

Variables Terminal

✓ 10:25 AM Python 3

Supernova remnant G46.8- | Problem loading page | Machine Learning e Inf | t.dvi - minka-matrix.pdf | raw.githubusercontent.com | numpy add column of | hadamard product - Bu | Home | Prop_algebra | Reg_Mult_G2.ipynb - C
colab.research.google.com/drive/1sH0aJZPyiKmhcemjfXjG7zuxZyOMM-25#scrollTo=LGEjf79ts9oF 180% VPN

Df Residuals:	196			**BIC:**	793.6	
Df Model:	3					
Covariance Type:	nonrobust					
	coef	**std err**	**t**	**P>|t|**	**[0.025**	**0.975]**
Intercept	2.9389	0.312	9.422	0.000	2.324	3.554
TV	0.0458	0.001	32.809	0.000	0.043	0.049
Radio	0.1885	0.009	21.893	0.000	0.172	0.206
Newspaper	-0.0010	0.006	-0.177	0.860	-0.013	0.011
Omnibus:	60.414	**Durbin-Watson:**	2.084			
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	151.241			

P. Hipotesis

$H_0, \beta = 0$

$H_1, \beta \neq 0$

Terminal

Supernova remnant G46.8- | Problem loading page | Machine Learning e Inf | t.dvi - minka-matrix.pdf | raw.githubusercontent.com | numpy add column of | hadamard product - Bu | Home | Prop_algebra | Reg_Mult_G2.ipynb
colab.research.google.com/drive/1sH0aJZPyiKmhcemjfXjG7zuxZyOMM-25#scrollTo=4yr40rO9tmmo
Commands | + Code | + Text | Run all

Method:	Least Squares	**F-statistic:**	859.6
Date:	Mon, 14 Sep 2026	**Prob (F-statistic):**	4.83e-98
Time:	16:36:54	**Log-Likelihood:**	-386.20
No. Observations:	200	**AIC:**	778.4
Df Residuals:	197	**BIC:**	788.3
Df Model:	2		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	2.9211	0.294	9.919	0.000	2.340	3.502
TV	0.0458	0.001	32.909	0.000	0.043	0.048
Radio	0.1880	0.008	23.382	0.000	0.172	0.204

Omnibus:	60.022	**Durbin-Watson:**	2.081
Prob(Omnibus):	0.000	**Jarque-Bera (JB):**	148.679
Skew:	-1.323	**Prob(JB):**	5.19e-33
Kurtosis:	6.292	**Cond. No.**	425.

Notes:

$\hat{y} = \beta_0 + \beta_1 x$

Recta

$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2$

Plano

Hiperplano

$Sales = (2.92) + (0.04)TV + (0.19)Radio$

Variables | Terminal | 10:36 AM